C++ Practice Questions

CSE141 Introduction to Programming

Instructions

- Write a separate C++ program for each question.
- Use only the concepts covered in lectures and labs.
- Read each question carefully and decide which C++ features are required.
- Unless otherwise stated, take input from the user where appropriate.
- Do not use loops, arrays, functions, or other topics that have not yet been covered.
- Questions are divided into three difficulty levels: **Easy**, **Medium**, and **Hard**.

Topic 1: Output and Escape Sequences

Question 1 *[Easy]*

Write a program that displays your name, university, and degree on three separate lines using `\n`.

Question 2 *[Easy]*

Write a program that displays a small menu using `\t` to align the item names and prices.

Question 3 *[Medium]*

Write a program that displays the following information using a single `cout` statement:

```
Name: Abdullah
Course: CSE141
Section: A
Status: "Enrolled"
```

Use appropriate escape sequences for the quotation marks and new lines.

Question 4 *[Medium]*

Write a program that displays a three-line message where each line is separated using `\n` and the values are aligned using `\t`.

Question 5 *[Hard]*

Write a program that displays a simple receipt using only one `cout` statement. Use `\n`, `\t`, and `\"` appropriately.

Topic 2: Variables

Question 6 ***[Easy]*** . . . .

Declare an integer variable called `age`, assign it a value, and display it.

Question 7 ***[Easy]*** . . . .

Declare variables for a student's name, age, and GPA. Assign suitable values and display them.

Question 8 ***[Medium]*** . . . .

Create variables to store the price of a product and its quantity. Display both values and calculate the total price.

Question 9 ***[Medium]*** . . . .

Create variables for a student's name, semester number, and number of completed courses. Display the information in a formatted student record.

Question 10 ***[Hard]*** . . . .

Create variables for the length and width of a rectangle. Calculate and display its area and perimeter.

Topic 3: Input and Output

Question 11 ***[Easy]*** . . . .

Ask the user for their name and display a greeting containing their name.

Question 12 ***[Easy]*** . . . .

Ask the user for two integers and display both values.

Question 13 ***[Medium]*** . . . .

Ask the user for the price and quantity of a product. Display the total cost.

Question 14 ***[Medium]*** . . . .

Ask the user for their name, age, and semester. Display the information as a student profile.

Question 15 ***[Hard]*** . . . .

Ask the user for the prices of three products. Display each price and the total amount to be paid.

Topic 4: Data Types

Question 16 *[Easy]*

Declare one variable of each of the following types: `int`, `double`, `char`, `bool`, and `string`. Assign suitable values and display them.

Question 17 *[Easy]*

Create a program that stores a student's age as an `int` and GPA as a `double`. Display both values.

Question 18 *[Medium]*

Create variables to store a product's name, price, quantity, and availability. Choose appropriate data types for each variable and display them.

Question 19 *[Medium]*

Create variables to store a person's name, age, height, and gender initial. Choose appropriate data types and display the information.

Question 20 *[Hard]*

Create variables for a student's roll number, percentage, grade character, and pass/fail status. Choose suitable data types and display all four values.

Topic 5: Mathematical Operations

Question 21 ***[Easy]*** . . . .

Write a program that takes two integers and displays their sum, difference, product, and quotient.

Question 22 ***[Easy]*** . . . .

Write a program that takes the length and width of a rectangle and calculates its area.

Question 23 ***[Medium]*** . . . .

Write a program that takes the radius of a circle and calculates its area using πr^2.

Question 24 ***[Medium]*** . . . .

Write a program that takes a temperature in Celsius and converts it to Fahrenheit using

$$F = \frac{9C}{5} + 32$$

Question 25 ***[Hard]*** . . . .

Write a program that takes the total marks and marks obtained by a student and calculates the student's percentage.

Topic 6: Logical and Relational Operations

Question 26 ***[Easy]*** . . . .

Take two integers as input and display whether the first number is greater than the second number.

Question 27 ***[Easy]*** . . . .

Take a student’s age as input and display whether the student is at least 18 years old.

Question 28 ***[Medium]*** . . . .

Take two integers and determine whether both numbers are positive using logical operators.

Question 29 ***[Medium]*** . . . .

Take a student’s marks and determine whether the marks are between 0 and 100 using logical operators.

Question 30 ***[Hard]*** . . . .

Take a student’s marks and attendance percentage. Determine whether the student satisfies both conditions required to pass:

- Marks are at least 50.
- Attendance is at least 75

Topic 7: If-Else

Question 31 ***[Easy]*** . . . .

Take an integer as input and display whether it is positive or negative.

Question 32 ***[Easy]*** . . . .

Take a student's marks as input and display 'Pass" if the marks are at least 50; otherwise display 'Fail".

Question 33 ***[Medium]*** . . . .

Take an integer and determine whether it is even or odd.

Question 34 ***[Medium]*** . . . .

Take a person's age and display whether they are a child, teenager, or adult using appropriate `if-else` statements.

Question 35 ***[Hard]*** . . . .

Take three integers and display the largest number using `if-else` statements.

Topic 8: Nested If-Else

Question 36 ***[Easy]*** . . . .

Take a student's marks. If the student has passed, check whether the marks are at least 80 and display 'Excellent" or 'Good".

Question 37 ***[Easy]*** . . . .

Take a person's age. If they are at least 18, check whether they are at least 60 and display an appropriate message.

Question 38 ***[Medium]*** . . . .

Take a username and password as input. First check the username. If it is correct, check the password and display the appropriate message.

Question 39 ***[Medium]*** . . . .

Take a student's marks. If the marks are passing, use another condition to determine whether the student receives a distinction.

Question 40 ***[Hard]*** . . . .

Take three integers. Use nested `if-else` statements to determine the largest number.

Topic 9: Ternary Operator

Question 41 ***[Easy]*** . . . .

Take an integer and use the ternary operator to display whether it is positive or negative.

Question 42 ***[Easy]*** . . . .

Take a student’s marks and use the ternary operator to display ‘Pass” or ‘Fail”.

Question 43 ***[Medium]*** . . . .

Take two integers and use the ternary operator to display the larger number.

Question 44 ***[Medium]*** . . . .

Take an integer and use the ternary operator to determine whether it is even or odd.

Question 45 ***[Hard]*** . . . .

Take a student’s marks and use a nested ternary operator to display:

- ‘Excellent” for marks 80 or above,
- ‘Good” for marks from 50 to 79,
- “Fail” for marks below 50.

Topic 10: Type Casting

Question 46 ***[Easy]*** . . . .

Take two integers and display their division as a decimal value using type casting.

Question 47 ***[Easy]*** . . . .

Create a `double` variable and convert it to an `int` using an appropriate casting method. Display both values.

Question 48 ***[Medium]*** . . . .

Take the marks obtained and total marks as integers. Calculate the percentage as a decimal value using type casting.

Question 49 ***[Medium]*** . . . .

Take a character containing a digit such as `'7'` and convert it to its corresponding integer value.

Question 50 ***[Hard]*** . . . .

Take a decimal number as input and display:

- its integer part,
- its value rounded to the nearest integer,
- and its original decimal value.

Use appropriate casting or mathematical functions where necessary.

Mixed Practice

These questions combine two or more of the topics covered above.

Question 51 ***[Easy]*** . . . .

Take two integers as input and display their sum. Then use an `if-else` statement to display whether the sum is even or odd.

Question 52 ***[Easy]*** . . . .

Take a student's name and marks as input. Display the name and use an `if-else` statement to display 'Pass" or 'Fail".

Question 53 ***[Medium]*** . . . .

Take the price and quantity of a product. Calculate the total price and use the ternary operator to display "Discount Applied" if the total is greater than or equal to 5000.

Question 54 ***[Medium]*** . . . .

Take three numbers as input. Use mathematical and relational operators to determine whether all three numbers are positive.

Question 55 ***[Medium]*** . . . .

Take a student's marks and attendance. Use nested `if-else` statements to determine whether the student is eligible to pass the course.

Question 56 ***[Medium]*** . . . .

Take two integers and calculate their average. Use type casting so that the answer is displayed as a decimal value.

Question 57 ***[Hard]*** . . . .

Take a student's marks and attendance percentage. Determine the student's result using the following rules:

- Marks must be at least 50 and attendance must be at least 75
- A student with marks of 80 or above receives 'A".
- Otherwise, a passing student receives 'B".

Use nested `if-else` statements.

Question 58 ***[Hard]*** . . . .

Take the prices of three products. Calculate the total price and apply the following rule:

- If the total is at least 10,000, apply a 10
- Otherwise, no discount is applied.

Display the original total, discount, and final amount.

Question 59 ***[Hard]*** . . . .

Take an integer as input. Use mathematical and logical operators along with `if-else` statements to determine whether the number is:

- positive and even,
- positive and odd,
- negative and even,
- or negative and odd.

Question 60 *[Hard]*

Take two integers as input. Display their quotient as a decimal value using type casting. Then use a ternary operator to display whether the quotient is greater than or equal to 1.

Optional Challenge Questions

Challenge 1 . . . .

Take three integers and determine the largest and smallest number using `if-else` statements.

Challenge 2 . . . .

Take a student's marks and display their grade using nested `if-else` statements.

Challenge 3 . . . .

Take the amount of a shopping bill and use the ternary operator to determine whether the customer receives a discount.

Challenge 4 . . . .

Take two integers and display their average as both an integer and a decimal value. Use type casting where required.

Challenge 5 . . . .

Take a person's age and determine whether they are eligible for a driving license. If they are eligible, use another condition to determine whether they are eligible for a senior discount.